

In our study, several different interaction forms are used to find the equilibrium states in a parabolic confining well at zero temperature. For the Yukawa-type interaction, the effect of changing Debye length λ_D is checked. $\lambda_D = \infty$ corresponds to the case of unshielded charged particles on a flat 2D surface. We also test the case with $V_{ij} = \ln(1/r_{ij})$, which is the interacting potential between two uniformly charged infinite long lines i and j to simulate the ideal 2D Coulomb interaction. Note that in the ideal 2D case (i.e., with a uniform axial charge distribution), the parabolic confining potential is equivalent to the field generated by a uniform frozen neutralizing ion background. The cases with a fourth power term added on the parabolic confining potential are also tested to check the effect of the steeper confining potential in the outer region of the clusters, which might occur for the case of confining charged dust particles by the space charge field in our experimental plasma system [16]. The degrees of lattice bending for forming a circular outer shell and the local adjustment to match the interface between the inner triangular domain and outer circular shells are manifested by the presence of the disclination defects with $+1$ and -1 topological charges (represented by squares and triangles, respectively, in the corresponding figures in this paper).

III. RESULTS AND DISCUSSIONS

A. Packing sequence at small N

For a cluster at small N , particles are packed into concentric shells. Table I shows the packing sequences of the ground states for several different interaction forms. The typical cluster structures are also shown in Figs. 2–4. The state $(N_1, N_2, N_3, \dots)$ corresponds to the state with occupation number N_i in the i th shell from the center. Increasing N one by one causes the variation of N_i . A new shell appears when the existing shells are full.

First we start from the example of the ideal 2D Coulomb system with $V_{ij} = \ln(1/r_{ij})$ in a parabolic V_c (Table I, type-IV interaction). For $N=3, 4$, and 5 , polygons are formed. For $N=6$, a particle appears at the center of the pentagon to minimize energy and forms the two-shell structure. Namely, a single shell structure with $N_1=6$ is not allowed because it costs too much confining energy. The states with $(1,6)$, $(1,7)$, $(1,8)$, and $(2,8)$ structures are found as N increases from 7 to 10 . Further increasing N to 16 causes the alternate increase of N_1 and N_2 until both shells are full (i.e., $N_1=5$ and $N_2=11$). For $N=17$, the third shell appears. It forms the $(1,5,11)$ structure. For the structure with three shells, similar filling processes are observed. The occupation number difference between the adjacent shells, $\Delta N_i = N_{i+1} - N_i$, usually varies between 5 and 6 . It can change to 4 or 7 around the transition of increasing one extra shell. Note that many nearly degenerate metastable states have also been observed (see the examples in Figs. 3 and 4).

For the cases of Yukawa-type interaction $V_{ij} = \exp(-r_{ij}/\lambda_D)/r_{ij}$, which has a shorter interaction range and harder repulsive core for each particle at small λ_D , similar packing results are obtained. The results from the three representative cases with $\lambda_D=0.1, 1$, and ∞ are presented in Table I (the type-I, -II, and -III interactions, respectively). The ground-state configurations for the case of $\lambda_D=\infty$ are

TABLE I. The Mendeleev table for the particle packing sequences from $N=1$ to 30 under different combinations of the confining and the mutual repulsion potentials. I to V are also used in other figures to represent the corresponding type of interaction and confinement. r_{cN} used in the type-V confining potential is the cluster radius obtained in the type-III interaction at the corresponding N .

V_c	r_i^2				$r_i^2 + r_i^4/r_{cN}^2$	
	V_{ij}	$\exp(-r_{ij}/\lambda_D)/r_{ij}$	$\ln(1/r_{ij})$	$1/r_{ij}$	Expt.	
N	$\lambda_D=0.1$	$\lambda_D=1.0$	$\lambda_D=\infty$			
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	1,5	1,5	1,5	1,5	6	1,5
7	1,6	1,6	1,6	1,6	1,6	1,6
8	1,7	1,7	1,7	1,7	1,7	2,6
9	2,7	2,7	2,7	1,8	1,8	2,7
10	2,8	2,8	2,8	2,8	2,8	2,8
11	3,8	3,8	3,8	3,8	2,9	3,8
12	3,9	3,9	3,9	3,9	3,9	3,9
13	4,9	4,9	4,9	4,9	3,10	4,9
14	4,10	4,10	4,10	4,10	4,10	4,10
15	5,10	5,10	5,10	4,11	4,11	5,10
16	1,5,10	1,5,10	1,5,10	5,11	5,11	5,11
17	1,6,10	1,6,10	1,6,10	1,5,11	5,12	1,5,11
18	1,6,11	1,6,11	1,6,11	1,6,11	1,5,12	1,6,11
19	1,6,12	1,6,12	1,6,12	1,6,12	1,6,12	1,6,12
20	1,7,12	1,7,12	1,7,12	1,7,12	1,6,13	1,7,12
21	2,7,12	2,7,12	1,7,13	1,7,13	1,7,13	1,7,13
22	2,8,12	2,8,12	2,8,12	1,7,14	1,7,14	2,7,13
23	3,8,12	3,8,12	2,8,13	1,8,14	1,8,14	2,8,13
24	3,8,13	3,8,13	3,8,13	2,8,14	2,8,14	2,8,14
25	3,9,13	3,9,13	3,9,13	3,8,14	2,8,15	3,8,14
26	4,9,13	4,9,13	3,9,14	3,9,14	3,9,14	3,9,14
27	4,9,14	4,9,14	4,9,14	3,9,15	3,9,15	3,9,15
28	4,10,14	4,10,14	4,10,14	4,9,15	3,9,16	4,9,15
29	4,10,15	5,10,14	4,10,15	4,10,15	3,10,16	4,10,15
30	5,10,15	5,10,15	5,10,15	4,10,16	4,10,16	4,10,16
	I	II	III	IV	V	VI

shown in Fig. 2. The harder repulsive core with the shorter-range interaction makes N_i deviate from those of the $\ln(1/r_{ij})$ interaction for certain cases. It tends to have a smaller cluster with the inner core, which has higher packing density. It might also change the ground-state configuration. For example, in the case of $V_{ij} = 1/r_{ij}$ (i.e., $\lambda_D = \infty$), the $(1,8)$ state which is the stable state for $V_{ij} = \ln(1/r_{ij})$ becomes metastable and the state $(2,7)$ becomes the new ground state. The old ground states $(5,11)$, $(1,7,14)$, $(1,8,14)$, $(3,8,14)$, $(3,9,15)$, $(4,9,15)$, and $(4,10,16)$ become metastable and are replaced by the new ground states $(1,5,10)$, $(2,8,12)$, $(2,8,13)$, $(3,9,13)$, $(4,9,14)$, $(4,10,14)$, and $(5,10,15)$, respectively. Namely, ΔN_i might decrease by one or two. Note that usually the energy differences between the above ground and metastable states are very small ($<0.05\%$). Table I shows that decreasing λ_D from ∞ to 0.1 tends to further decrease ΔN_i by one for some configurations.